ROCSUR
Rocuronium Bromide Injection
10 mg/mL, 5mL Vial

Qualitative and quantitative composition
Rocuronium Bromide USP......10 mg
Water for Injection USP q.s. to 1 mL

Pharmaceutical form
Rocuronium Bromide Injection is supplied as an aqueous solution for intravenous injection.

Description:
Clear, colourless to yellow or orange solution.

Pharmacological properties

1. Pharmacodynamic properties
Rocuronium bromide is a fast onset, intermediate acting non-depolarizing neuromuscular blocking agent, possessing all of the characteristic pharmacological actions of this class of drugs (curariform). It acts by competing for nicotinic cholinoceptors at the motor end-plate. This action is antagonized by acetylcholinesterase inhibitors such as neostigmine, edrophonium and pyridostigmine.

The ED90 (dose required to produce 90% depression of the twitch response of the thumb to stimulation of the ulnar nerve) during intravenous anesthesia is approximately 0.3 mg rocuronium bromide per kg body weight.

The ED95 in infants is lower than in adults and children (0.25, 0.35 and 0.40 $mg.kg^{-1}$, respectively).

Within 60 seconds following intravenous administration of a dose of 0.6 mg Rocuronium bromide per kg body weight (2 x ED90 under intravenous anesthesia), adequate intubation conditions can be achieved in nearly all patients of which in 80% intubation conditions are rated excellent. General muscle paralysis adequate for any type of procedure is established within 2 minutes. The clinical duration (the duration until spontaneous recovery to 25% of control twitch height) with this dose is 30-40 minutes. The total duration (time until spontaneous recovery to 90% of control twitch height) is 50 minutes. The mean time of spontaneous recovery of twitch response from 25 to 75% (recovery index) after a bolus dose of 0.6 mg rocuronium bromide per kg body weight is 14 minutes.

With lower dosage of 0.3-0.45 mg rocuronium bromide per kg body weight (1-1½ x ED90), onset of action is slower and duration of action is shorter. After administration of 0.45 mg rocuronium bromide per kg body weight, acceptable intubation conditions are present after 90 seconds. During rapid sequence induction of anesthesia under propofol or fentanyl/thiopental anesthesia, adequate intubation conditions are achieved within 60 seconds in 93% and 96% of the patients respectively, following a dose of 1.0 mg rocuronium bromide per kg body weight. Of these, 70% are rated excellent. The clinical duration with this dose approaches 1 hour, at which time the neuromuscular block can be safely reversed. Following a dose of 0.6 mg rocuronium bromide per kg body weight, adequate intubation conditions are achieved within 60 seconds in 81% and 75% of the patients during a rapid sequence induction technique with propofol or fentanyl/thiopental, respectively.

The duration of action of maintenance doses of 0.15 mg rocuronium bromide per kg body weight might be somewhat longer under enflurane and isoflurane anesthesia in geriatric patients and in patients with hepatic disease and/or renal disease (approximately 20 minutes) than in patients without impairment of excretory organ functions under intravenous anesthesia (approximately 13 minutes). No cumulation of effect (progressive increase in duration of action) with repetitive maintenance dosing at the recommended level has been observed.

Following continuous infusion in the Intensive Care Unit, the time to recovery of the train of four ratios to 0.7 depends on the level of block at the end of the infusion. After a continuous infusion for 20 hours or more the median (range) time between return of T2 to train of four stimulation and recovery of the train of four to ratio to 0.7 approximates 1.5 (1-5) hours in patients without multiple organ failure and 4(1-25) hours in patients with multiple organ failure.

In patients scheduled for cardiovascular surgery the most common cardiovascular changes during the onset of maximum block following 0.6-0.9 mg Rocuronium Bromide per kg body weight are a slight and clinically insignificant increase in heart rate up to 9% and an increase in mean arterial blood pressure up to 16% from the control values. Administration of acetylcholinesterase inhibitors, such as neostigmine, pyridostigmine or edrophonium, antagonizes the action of Rocuronium Bromide.

Special populations:
Mean onset time in infants and children at an intubation dose of 0.6 mg/kg body weight is slightly shorter than in adults. The duration of relaxation and the time to recovery tend to be shorter in children compared to infants and adults.

Pharmacokinetic properties
After intravenous administration of a single bolus dose of rocuronium bromide the plasma concentration time course runs in three exponential phases. In normal adults, the mean (95% CI) elimination half-life is 73 (66-80) minutes, the (apparent) volume of distribution at steady state conditions is 203 (193-214) $ml.kg^{-1}$ and plasma clearance is 3.7 (3.5-3.9) $ml.kg^{-1}.min^{-1}$. In controlled studies the plasma clearance in geriatric patients and in patients with renal dysfunction was reduced, in most studies however without reaching the level of statistical significance. In patients with hepatic disease, the mean elimination half-life is prolonged with 30 minutes and the mean plasma clearance is reduced with 1 $ml.kg^{-1}.min^{-1}$.

When administered as a continuous infusion to facilitate mechanical ventilation for 20 hours or more, the mean elimination half-life and the mean (apparent) volume of distribution at steady state are increased. A large between patient variability is found in controlled clinical studies, related to nature and extent of (multiple) organ failure and individual patient characteristics. In patients with multiple organ failure a mean (± SD) elimination half-life of 21.5 (± 3.3) hours, a (apparent) volume of distribution at steady state of 1.5 (± 0.8) $ml.kg^{-1}$ and a plasma clearance of 2.1 (± 0.8) $ml.kg^{-1}.min^{-1}$ were found.

Rocuronium is excreted in urine and bile. Excretion in urine approaches 40% within 12-24 hours. After injection of radiolabel is on average 47% in urine and 43% in feces after 9 days. Approximately 50% is recovered as the parent compound.

Indications
Rocuronium Bromide Injection is indicated as an adjunct to general anesthesia to facilitate endotracheal intubation, to provide skeletal muscle relaxation and to facilitate mechanical ventilation in adults, children and infants from one month of age. Rocuronium Bromide Injection is also indicated as an adjunct in the intensive care unit (ICU) to facilitate mechanical ventilation and as part of Rapid Sequence Induction. However, this has not been studied in infants and children.

Recommended dosage
As with other neuromuscular blocking agents, the dosage of Rocuronium Bromide Injection should be individualized in each patient. The method of anesthesia and the expected duration of surgery, the method of sedation and the expected duration of mechanical ventilation, the possible interaction with other drugs that are administered concomitantly, and the condition of the patient should be taken into account when determining the dose. The use of an appropriate neuromuscular monitoring technique is recommended for the evaluation of neuromuscular block and recovery.

Inhalational anesthetics do potentiate the neuromuscular blocking effects of Rocuronium Bromide Injection. This potentiation however, becomes clinically relevant in the course of anesthesia, when the volatile agents have reached the tissue concentrations required for this interaction. Consequently, adjustments with Rocuronium Bromide Injection should made by administering smaller maintenance doses at less frequent intervals or by using lower infusion rates of Rocuronium Bromide Injection during long lasting procedures (longer than 1 hour) under inhalational anesthesia (see Interaction with other medicaments and other forms of interaction).

In adult patients the following dosage recommendations may serve as a general guideline for tracheal intubation and muscle relaxation for short to long lasting surgical procedures and for use in the intensive care unit.

Surgical Procedures

Tracheal intubation:

The standard intubating dose during routine anesthesia is 0.6 mg rocuronium bromide per kg body weight, after which adequate intubation conditions are established within 60 seconds in nearly all patients. A dose of 1.0 mg rocuronium bromide per kg body weight is recommended for facilitating tracheal intubation conditions during rapid sequence induction of anesthesia, after which adequate intubation conditions are also established within 60 seconds in nearly all patients. If a dose of 0.6 mg rocuronium bromide per kg body weight is used for rapid sequence induction of anesthesia, it is recommended to intubate the patient 90 seconds after administration of rocuronium bromide. In patients undergoing Cesarean section it is recommended to only use a dose of 0.6 mg rocuronium bromide per kg body weight, since a 1.0 mg/kg dose has not been investigated in this patient group.

Maintenance dosing:
The recommended maintenance dose is 0.15 mg rocuronium bromide per kg body weight; in the case of long-term inhalational anesthesia, this should be reduced to 0.075-0.1 mg rocuronium bromide per kg body weight. The maintenance doses should best be given when twitch height has recovered to 25% of control twitch height, or when 2 to 3 responses to train of four stimulations are present.

Continuous infusion:
If Rocuronium Bromide Injection is administered by continuous infusion, it is recommended to give a loading dose of 0.6 mg rocuronium bromide per kg body weight and, when neuromuscular block starts to recover, to start administration by infusion. The infusion rate should be adjusted to maintain twitch response at 10% of control twitch height or to maintain 1 to 2 responses to train of four stimulations. In adults under intravenous anesthesia, the infusion rate required to maintain neuromuscular block at this level ranges from 0.3-0.6 $mg.kg^{-1}.h^{-1}$ and under inhalational rate ranges from 0.3-0.4 $mg.kg^{-1}.h^{-1}$. Continuous monitoring of neuromuscular block is recommended since infusion rate requirements vary from patient to patient and with the anesthetic method used.

Dosing in geriatric patients and patients with hepatic and/or biliary tract disease and/or renal failure:
The standard intubation dose for geriatric patients and patients with hepatic and/or biliary tract disease and/or renal failure during routine anesthesia is 0.6 mg rocuronium bromide per kg body weight. A dose of 0.6 mg per kg body weight should be considered for rapid sequence induction of anesthesia in patients in which a prolonged duration of action is expected. Regardless of the anesthetic technique used, the recommended maintenance dose for these patients is 0.075-0.1 mg rocuronium bromide per kg body weight, and the recommended infusion rate is 0.3-0.4 $mg.kg^{-1}.h^{-1}$ (see also continuous infusion).

Dosing in overweight and obese patients:
When used in overweight or obese patients with a body weight of 30% or more above ideal body weight) doses should be reduced taking into account a lean body mass.

Intensive Care Procedures

Tracheal intubation
For tracheal intubation, the same doses should be used as described above under surgical procedures.

Dosing to Facilitate Mechanical Ventilation
The use of an initial loading dose of 0.6 mg rocuronium bromide per kg body weight is recommended, followed by a continuous infusion as soon as twitch height recovers to 10% or upon reappearance of 1 to 2 twitches to train of four stimulations. Dosage should always be titrated to effect in the individual patient. The recommended initial infusion rate for the maintenance of a neuromuscular block of 80-90% (1 to 2 twitches to TOF stimulation) in adult patients is 0.3-0.6 $mg.kg^{-1}.h^{-1}$ during the first hour of administration, which will need to be decreased during the following 6-12 hours, according to the individual response. Thereafter, individual dose requirements remain relatively constant.

A large between patient variability in hourly infusion rates has been found in controlled clinical studies, with mean hourly infusion rates ranging from 0.2-0.5 $mg.kg^{-1}.h^{-1}$ depending on nature and extent of organ failure(s), concomitant medication and individual patient characteristics. To provide optimal individual patient control, monitoring of neuromuscular transmission is strongly recommended. Administration up to 7 days has been investigated. There are no data to support dose recommendations for the facilitation of mechanical ventilation in pediatric and geriatric patients.

Administration
Rocuronium Bromide Injection is administered intravenously either as a bolus injection or as a continuous infusion (see also Instructions for use/handling).

Contra-indications
Former anaphylactic reactions to rocuronium or to the bromide ion.

Special warnings and special precautions for use
Since Rocuronium Bromide Injection causes paralysis of the respiratory muscles, ventilator support is mandatory for patients treated with this drug until adequate spontaneous respiration is restored. As with all neuromuscular blocking agents, it is important to anticipate intubation difficulties, particularly when used as part of a rapid sequence induction technique. Anaphylactic reactions can occur following the administration of neuromuscular blocking agents. Precautions for treating such reactions should always be taken. Particularly in the case of previous anaphylactic reactions to neuromuscular blocking agents, special precautions should be taken since allergic cross-reactivity to neuromuscular blocking agents has been reported.

In general, following long term use of neuromuscular blocking agents in the ICU, prolonged paralysis and/or skeletal muscle weakness has been noted. In order to help preclude possible prolongation of neuromuscular block and/or over dosage it is strongly recommended that neuromuscular transmission is monitored throughout the use of muscle relaxants. In addition, patients should receive adequate analgesia and sedation. Furthermore, muscle relaxants should be titrated to effect in the individual patients by or under supervision of experienced clinicians who are familiar with their actions and with appropriate neuromuscular monitoring techniques.

The following conditions may influence the pharmacokinetics and/or pahrmacodynamics of Rocuronium Bromide Injection:

Hepatic and/or biliary tract disease and renal failure
Because Rocuronium is excreted in urine and bile, Rocuronium Bromide Injection should be used with caution in patients with clinically significant hepatic and/or biliary diseases and/or renal failure. In these patient groups prolongation of action has been observed with doses of 0.6 mg Rocuronium bromide per kg body weight.

Prolonged circulation time
Conditions associated with prolonged circulation time such as cardiovascular disease, old age and oedematous state resulting in an increased volume of distribution, may contribute to a slower onset of action.

Neuromuscular disease
Like other neuromuscular blocking agents, Rocuronium Bromide Injection should be used with extreme caution in patients with a neuromuscular disease or after poliomyelitis since the response to neuromuscular blocking agents may be considerably altered in these cases. The magnitude and direction of this alteration may vary widely. In patients with myasthenia gravis or with the myasthenic (Eaton-Lambert) syndrome, small doses of Rocuronium Bromide Injection may have profound effects and Rocuronium Bromide Injection should be titrated to the response.

Hypothermia
In surgery under hypothermic conditions, the neuromuscular blocking effect of Rocuronium Bromide Injection is increased and the duration prolonged.

Obesity
Like other neuromuscular blocking agents, Rocuronium Bromide Injection may exhibit a prolonged duration and a prolonged duration and a prolonged spontaneous recovery in obese patients, when the administered doses are calculated on actual body weight.

Burns
Patients with burns are known to develop resistance to non-depolarizing neuromuscular blocking agents. It is recommended that the dose is titrated to response.

Conditions which may increase the effects of Rocuronium Bromide Injection
Hypokalaemia (e.g. after severe vomiting, diarrhea and diuretic therapy), hypermagnesemia, hypocalcemia (after massive transfusions), hypoproteinemia, dehydration, acidosis, hypercapnia, cachexia. Severe electrolyte disturbances, altered blood pH or dehydration should therefore be corrected when possible.

Interactions with other medicaments and other forms of interaction
The following drugs have been shown to influence the magnitude and/or duration of action of non-depolarizing neuromuscular blocking agents.

Increased effect
- Halogenated volatile anesthetics potentiate the neuromuscular block of Rocuronium Bromide Injection. The effect only becomes apparent with maintenance dosing. Reversal of the block with anticholinesterase inhibitors could also be inhibited.
- After intubation with suxamethonium.
- Long-term concomitant use of corticosteroids and Rocuronium Bromide Injection in the ICU may result in prolonged duration of neuromuscular block or myopathy. Other drugs.
- Antibiotics: aminoglycoside, lincosamide and polypeptide antibiotics, acylamino-penicillin antibiotics.
- Diuretics, quinidine and its isomer quinine, magnesium salts, calcium channel blocking agents, lithium salts, local anesthetics (lidocaine i.v, bupivacaine epidural) and acute administration of phenytoin or ß-blocking agents. Recurarization has been reported after post-operative administration of: aminoglycoside, lincosamide, polypeptide and acylamino-penicillin antibiotics, quinidine, quinine and magnesium salts.

Decreased effect
- Prior chronic administration of phenytoin or carbamazepine.
- Protease inhibitors (gabexate, linastatin)

Variable effect
- Administration of other non-depolarizing neuromuscular blocking agents in combination with Rocuronium Bromide Injection may produce attenuation or potentiation of the neuromuscular block, depending on the order of administration and the neuromuscular blocking agent used.
- Suxamethonium given after the administration of Rocuronium Bromide Injection may produce potentiation or attenuation of the neuromuscular blocking effect of Rocuronium Bromide Injection.

Effect of Rocuronium Bromide Injection on other drugs:
Rocuronium Bromide Injection combined with lidocaine may result in a quicker onset of action of lidocaine.

Pregnancy and lactation

Pregnancy:
Caution should be exercised when prescribing Rocuronium Bromide Injection to pregnant women.

Cesarean section:
In patients undergoing Cesarean section, Rocuronium Bromide Injection can be used as part of a rapid sequence induction technique, provided no intubation difficulties are anticipated and a sufficient dose of anesthetic agent is administered or following suxamethonium facilitated intubation. Rocuronium Bromide Injection, administered in doses of 0.6 mg/kg, has been shown to be safe in parturients undergoing Cesarean section. Rocuronium Bromide Injection does not affect Apgar score, fetal muscle tone nor cardiorespiratory adaptation. From umbilical cord blood sampling it is apparent that only limited placental transfer of rocuronium bromide occurs which does not lead to the observation of clinical adverse effects in the newborn.

Note 1: doses of 1.0 mg/kg have been investigated during rapid sequence induction of anesthesia, but not in Cesarean section patients. Therefore, only a dose of 0.6 mg/kg is recommended in this patient group.

Note 2: reversal of neuromuscular block induced by neuromuscular blocking agents may be inhibited or unsatisfactory in patients receiving magnesium salts for toxemia of pregnancy because magnesium salts enhance neuromuscular blockade. Therefore, in these patients the dosage of Rocuronium Bromide Injection should be reduced and be titrated to twitch response.

Lactation:
Rocuronium Bromide Injection should be given to lactating women only when the attending physician decides that the benefits outweigh the risks.

Side effects

Anaphylactic reactions
Although very rare, severe anaphylactic reactions to neuromuscular blocking agents, including Rocuronium Bromide injection, have been reported.

Anaphylactic/anaphylactoid reactions are: bronchospasm, cardiovascular changes (e.g. hypotension, tachycardia, circulatory collapse - shock), and cutaneous changes (e.g. angioedema, urticaria). These reactions have, in some cases, been fatal. Due to the possible severity of these reactions, one should always assume they may occur and take the necessary precautions.

Histamine release and histaminoid reactions
Since neuromuscular blocking agents are known to be capable of inducing histamine release both locally and systemically, the possible occurrence of itching and erythematous reactions at the site of injection and/or generalized histaminoid (anaphylactoid) reactions such as bronchospasm and cardiovascular changes e.g. hypotension and tachycardia should always be taken into consideration when administering these drugs. In clinical studies only a slight increase in mean plasma histamine levels has been observed following rapid bolus administration of 0.3-0.9 mg rocuronium bromide per kg body weight.

Local injection site reactions
During rapid sequence induction of anesthesia, pain on injection has been reported, especially when the patient has not yet completely lost consciousness and particularly when propofol is used as the induction agent. In clinical studies, pain on the injection has been noted in 16% of the patients who underwent rapid sequence induction of anesthesia with propofol and in less than 0.5% of the patients who underwent rapid sequence induction of anesthesia with fentanyl and thiopental.

Overdosage
In the event of over dosage and prolonged neuromuscular block, the patient should continue to receive ventilator support and sedation. Upon start of spontaneous recovery an acetylcholinesterase inhibitor (e.g. neostigmine, edrophonium, pyridostigmine) should be administered in adequate doses. When administration of an acetycholinesterase inhibiting agent fails to reverse the neuromuscular effects of Rocuronium Bromide Injection, ventilation must be continued until spontaneous breathing is restored. Repeated dosage of an acetycholinesterase inhibitor can be dangerous.

In animal studies, severe depression of cardiovascular function, eventually leading to cardiac collapse did not occur until a cumulative dose of 750 x ED 90(135 mg rocuronium bromide per kg body weight) was administered.

Incompatibilities
Physical incompatibility has been documented for Rocuronium Bromide Injection when added to solutions containing the following drugs: amphotericin, amoxicillin, azathioprine, cefazolin, cloxacillin, dexamethasone, diazepam, enoximone, erythromycin, famotidine, frusemide, hydrocortisone sodium succinate, insulin, methohexital, methylprednisolone, prednisolone sodium succinate, thiopental, trimethoprim and vancomycin. Rocuronium Bromide is also incompatible with Intralipid.

Shelf life
Rocuronium Bromide Injection has a shelf life of 2 years. It will be stable for 6 months when stored below 25°C. Use the opened vial of Rocuronium Bromide injection within 30 days, if the vials are stored at a temperature between 20°- 25°C.

Special precautions for storage
Rocuronium Bromide Injection should be stored at 2° to 8°C in the dark. it will be stable for 6 months when stored below 25°C. Opened vial can be used within 30 days, if stored at temperature between 20° - 25°C.

Nature and contents of container
Rocuronium Bromide Injection 10 mg / ml are packed in USP type-I clear glass vial, each vial is packed in a mono or 10's carton with a package insert.

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